

UNIVERSITY OF ENGINEERING AND TECHNOLOGY, LAHORE

EDUMS

Education Management System

CSC-104L Database Systems Lab

Group DBS25F014

2024-CS-186	Muhammad Farhan Ali
2024-CS-212	Ali Hassaan
2024-CS-217	Amaan Sabir



Submitted To : Sir Nazeef Ul Haq

Section : D

Overview

Description

This project is made to tackle problems of students and teachers in an educational institute to tackle their problems. It involves solving problems of students and teachers and effectively managing their courses and assessments and results. It focuses on effective educational system to provide ease to both teacher and student to manage a course. This project is selected as final project for DB course as it includes many functionalities that covers our course. It will help students and teachers to easily use the application. Based on the user each one is provided a different interface to solve their desired queries and to perform specific tasks. In this teacher are assigned course or can ask for a course and then the student enrolls in that course and is graded by the teacher and so on and at the end his result is taken out. This application helps to manage the education system more easily in a more effective way.

Business Case

This application can be used by educational institutes to manage their students and teachers and course management to enforce smooth workflow in the institute and easily help the institute and the users to manage their data.

Motivation

Our educational website eduko motivated us to create an application version of it to replace the paperwork and other system to digital application system just like eduko. This motivated us to create EDUMS to counter education management issues just like eduko.

Audience

It targets teachers and students of educational institutes for easy education and course management and effective education.

Features

- User Friendly UI
- Easy Account Handling
- Student Attendance
- Course Management
- Maintains Results of Students
- Manage Student Assessments
- Easy Lecture Management
- Generate PDF reports by Admin

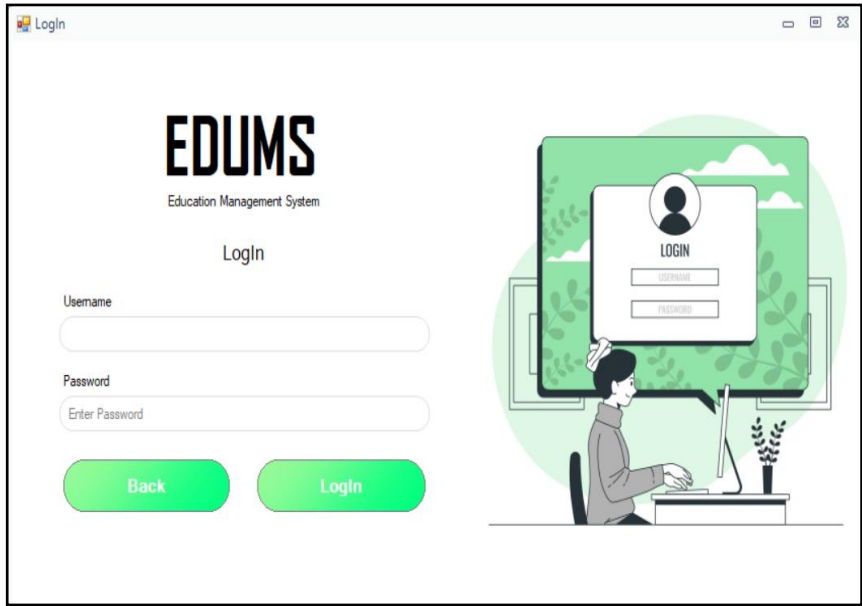
Technology Stack

Following technologies are used in this application

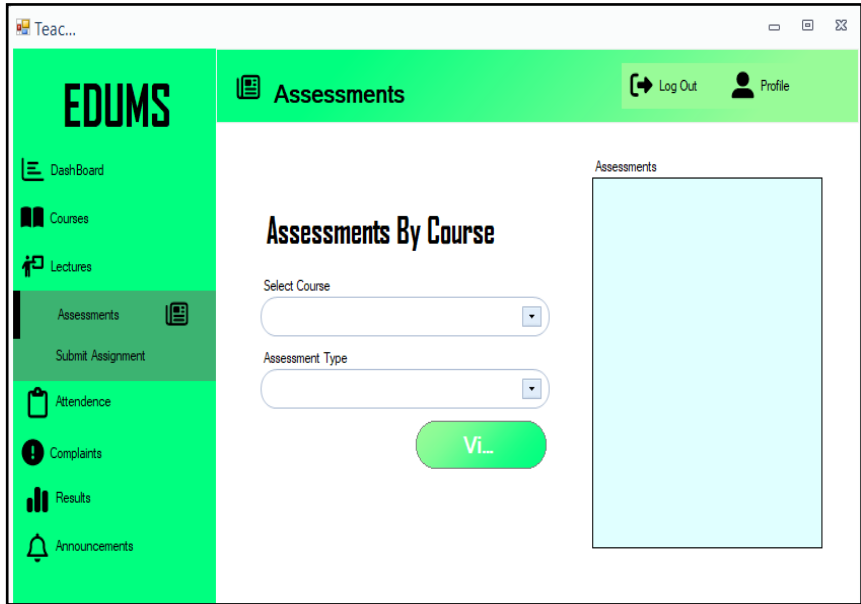
Platform	Desktop
Language	C#
Database	MySQL Database
Additional Packages	Krypton Forms for UI and ODBC for connection with database for reports

Wireframes/Use Cases

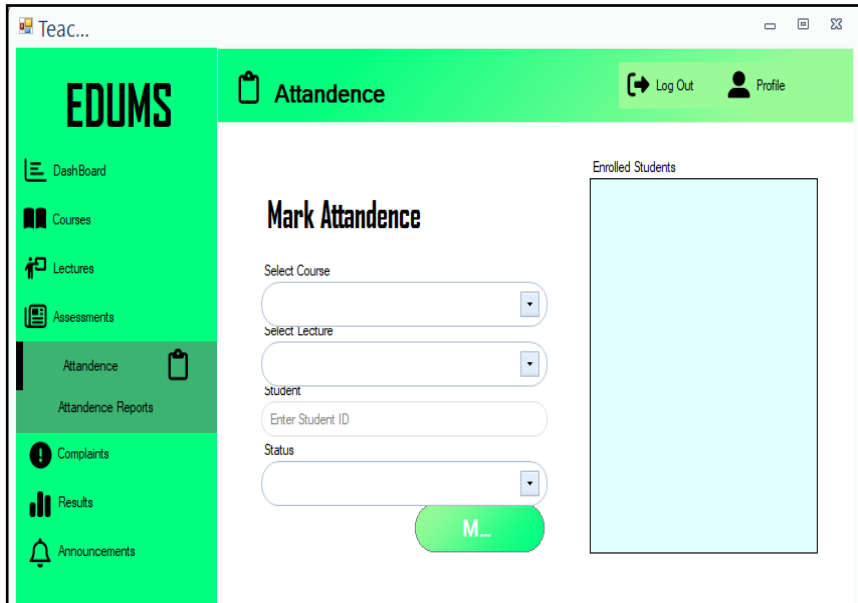
1. Login Page

Name	Login Page
Actor	User
Description	User Logs into the application according to his role and when he logs in the interface is shown to him according to his role and based on student, teacher or admin he can perform different functionalities and this is the page which decides which interface to show based on user and his role. Otherwise in case of incorrect user it doesn't log in.
UI	 The wireframe shows a web browser window titled 'Login'. The main heading is 'EDUMS' with the subtitle 'Education Management System'. Below this is a 'Login' section with two input fields: 'Username' and 'Password'. The 'Password' field has a placeholder text 'Enter Password'. At the bottom of the login section are two green buttons: 'Back' and 'Login'. To the right of the login section is an illustration of a person sitting at a desk with a computer, looking at a screen that displays a 'LOGIN' form with fields for 'USERNAME' and 'PASSWORD'.

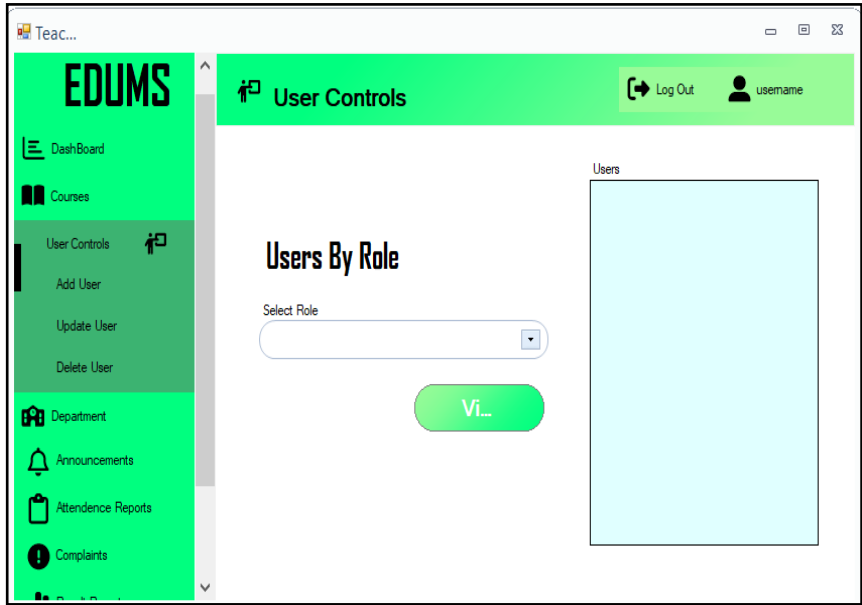
2. Assessment by Course

Name	Student Assessment by Course Page
Actor	User (Student)
Description	The student can submit Assessments/Assignments according to the course. He can submit the assignment assigned of a particular course using the easy interface of EDUMS application.
UI	

3. Mark Attendance

Name	Teacher Mark Attendance
Actor	User (Teacher)
Description	The teacher can easily mark attendance of a student in an enrolled course and mark him present or absent based on the status of student in the class.
UI	 The screenshot displays the 'Mark Attendance' interface within the EDUMS application. On the left is a green sidebar with navigation links: Dashboard, Courses, Lectures, Assessments, Attendance (highlighted), Attendance Reports, Complaints, Results, and Announcements. The main content area has a green header with 'Attendance' and 'Log Out'/'Profile' buttons. Below the header, the 'Mark Attendance' section includes dropdowns for 'Select Course' and 'Select Lecture', text inputs for 'Student' and 'Enter Student ID', and a 'Status' dropdown. A green 'Mark' button is at the bottom. To the right is a large light blue box labeled 'Enrolled Students'.

4. User Management

Name	Admin User Management
Actor	User (Admin)
Description	The admin can add users and delete and update them and also view them by role in the grid view given in the side. Admin will have the control to add other admins and teachers as well as students.
UI	

5. Attendance Report

Name	Attendance Report																		
Actor	User (Admin)																		
Description	The admin can view the attendance of students and can generate a report of it for based on course and lecture. This helps the admin to view attendance of students in an easy way.																		
UI And PDF Report	Teacher EDUMS Attendance Reports Log Out username Dashboard Courses User Controls Department Announcements Attendance Reports Report By Student Complaints Result Reports Student Report Teacher Reports Attendance By Lecture Select Course PF-102 Select Lecture intro View Attendance Report 1 of 1 100% <table><thead><tr><th>Student Name</th><th>Lecture</th><th>Status</th></tr></thead><tbody><tr><td>amaan sabir</td><td>intro</td><td>Present</td></tr><tr><td>ali</td><td>intro</td><td>Absent</td></tr></tbody></table> <table><thead><tr><th>Student Name</th><th>Lecture</th><th>Status</th></tr></thead><tbody><tr><td>amaan sabir</td><td>intro</td><td>Present</td></tr><tr><td>ali</td><td>intro</td><td>Absent</td></tr></tbody></table>	Student Name	Lecture	Status	amaan sabir	intro	Present	ali	intro	Absent	Student Name	Lecture	Status	amaan sabir	intro	Present	ali	intro	Absent
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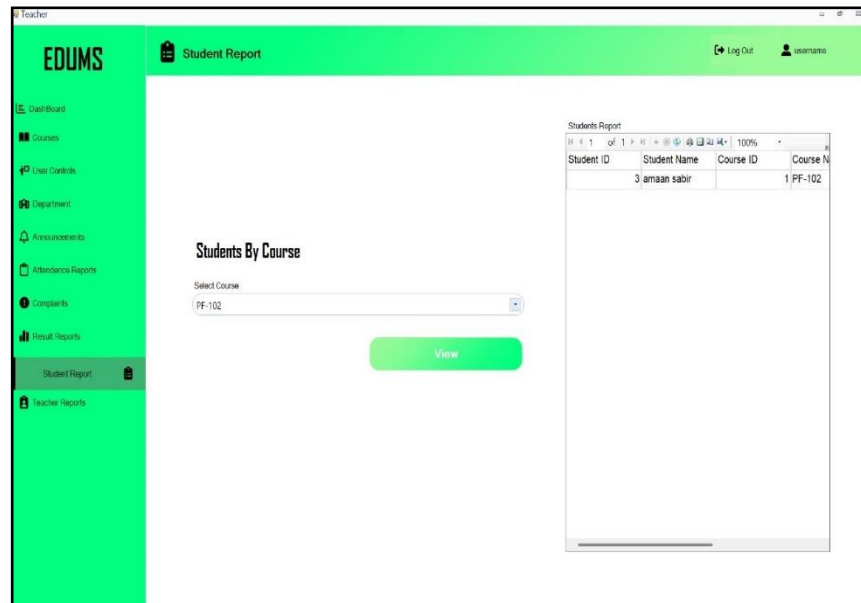
6. Teacher Report

Name	Teacher Report																				
Actor	User (Admin)																				
Description	To see all the teachers in the system the admin can generate a report of all the teachers in the system and can see the teachers registered in the education system.																				
UI And PDF Report	Teacher EDUMS Dashboard Courses User Controls Department Announcements Attendance Reports Complaints Result Reports Student Report Teacher Reports Teacher Reports Log Out username All Teachers 1 of 1 <table><thead><tr><th>Teacher ID</th><th>Name</th><th>Experience</th><th>Studies</th><th>Subject</th></tr></thead><tbody><tr><td>1</td><td>ali</td><td>3</td><td>cs</td><td>db</td></tr></tbody></table> <table><thead><tr><th>Teacher ID</th><th>Name</th><th>Experience</th><th>Studies</th><th>Subject</th></tr></thead><tbody><tr><td>1</td><td>ali</td><td>3</td><td>cs</td><td>db</td></tr></tbody></table>	Teacher ID	Name	Experience	Studies	Subject	1	ali	3	cs	db	Teacher ID	Name	Experience	Studies	Subject	1	ali	3	cs	db
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7. Student Enrollment Report

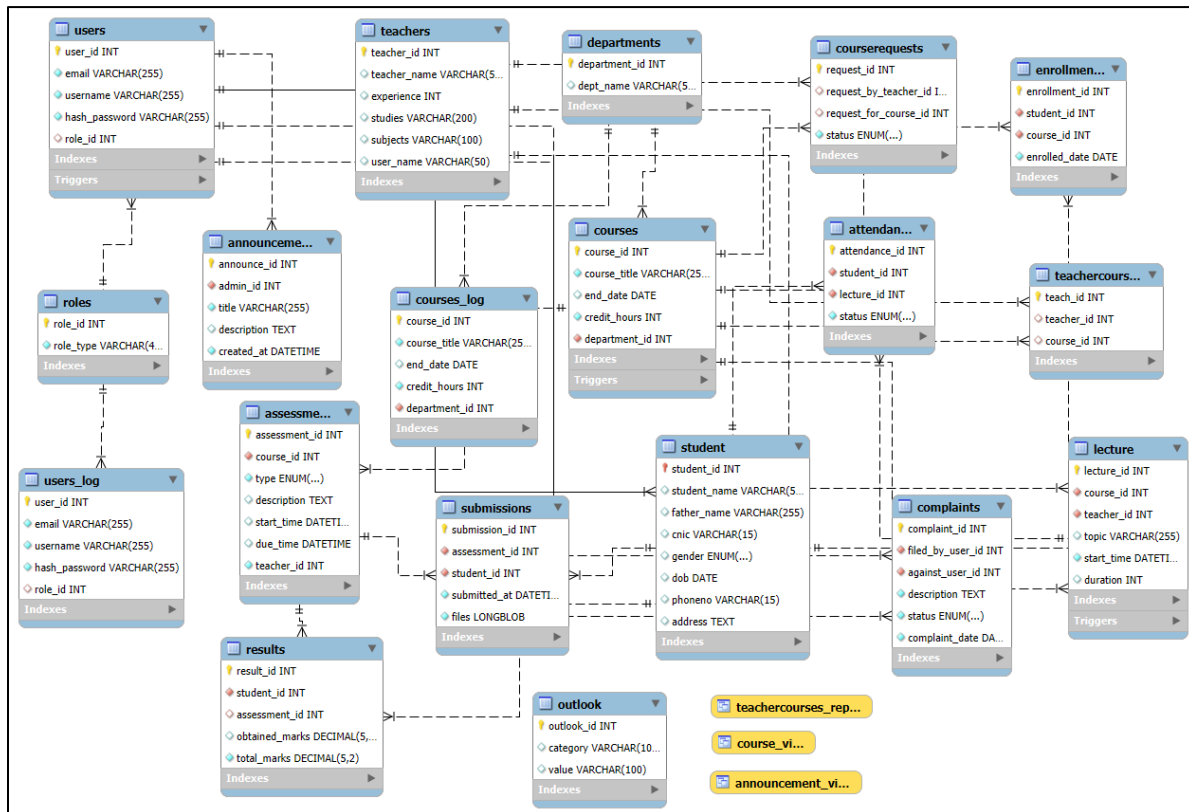
Name	Student Report
Actor	User (Admin)
Description	The admin can view the students enrolled in a course and can generate a report of the students enrolled in the course.

UI And
PDF Report



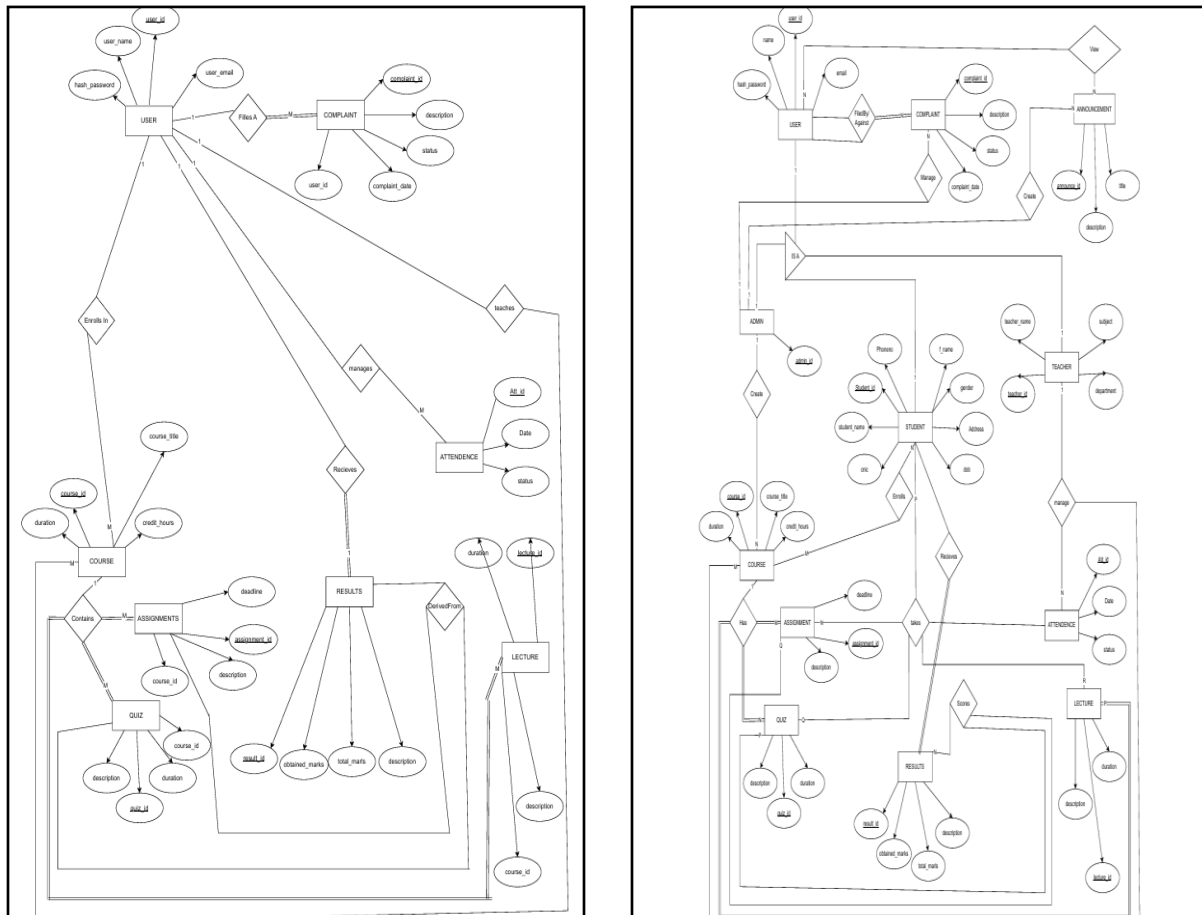
Student ID	Student Name	Course ID	Course Name
3	amaan sabir	1	PF-102

Database Design



Tables in database

- | | |
|---|--|
| <ul style="list-style-type: none"> • announcements • assessments • attendance • complaints • courserquests • courses • departments • enrollments • lecture | <ul style="list-style-type: none"> • outlook • results • roles • student • submissions • teachercourses • teachers • users |
|---|--|

ERD**Users of Application**

- **Student:** The student will be the main goal of this application as this mostly revolves around the user as he will be the one to enroll in courses and submit assignments and take lectures and son on.
- **Teacher:** The teacher will be the main supporting character as he will be the one who will create assignments and grade the student and teach a course.
- **Admin:** Admin will be the character managing the application as adding departments and new courses and managing both teacher and students. He will create reports and will have key role in the application.

Functional Requirements

	<i>As a</i>	<i>I want to perform</i>	<i>So that I can</i>
1	Student	Create an account Log into his account Register into a course Take lectures Submit assignments See results File complaints See Announcements View attendance	To use the app To see student interface To access that course To learn To get grades To see the results of assignments To complain against any issue To see any notifications To view his running attendance
2	Teacher	Create an account Log into his account Request Course Manage Assignments Manage Lectures See Announcements Make Results	To use the app To see teacher interface To teach a course To give assignments to student To give lectures to student To view notifications To grade the student
3	Admin	Create an account Log into his account Manage courses Manage Departments Manage Announcements See Attendance, Teacher, student, Complaint, Result reports	To use the app To see admin interface To create courses To manage departments To create announcements To view reports of different kinds

Data Structures

- **Lists:** Lists are used for storing objects as well as name of some attributes in some tables to extract ID from them.
- **Data Sources:** Data sources have been used for combo boxes and grid views.

Classes

